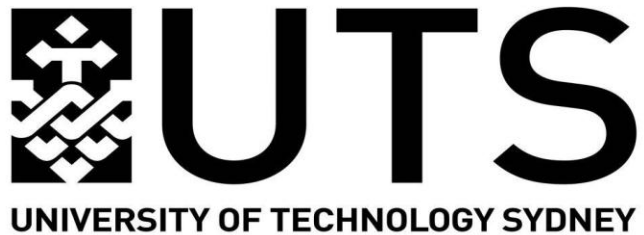


UNIVERSITY OF TECHNOLOGY SYDNEY



32146 Data Visualisation and Visual Analytics - Autumn 2025

Assessment Task 2: Advanced Data Visualisation

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Australia - 2025

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I. Dataset Summary and Trends

1.1 Dataset Overview

The dataset utilized in this project provides a rich historical record of the Australian Open tennis championships, spanning over 120 years of competition (from 1905 to 2025). It focuses on the men's and women's singles finals, offering detailed information on each championship match, including player names, nationalities, match scores, durations, seedings, and game-level statistics.

The dataset is structured across three Excel sheets:

- **Sheet 1** contains the core dataset of the Australian Open, with 212 records representing 210 final matches in both men's and women's singles.
- **Sheet 2** includes the pivot table used to calculate the top-performing players who won five or more based on the data from Sheet 1.
- **Sheet 3** presents the final list of top players who have secured five or more championship titles, comprising a total of 51 unique players.

1.2 Data exploration

- After exploring the dataset, there are some findings mentioned below:
- In 1966, the Score column indicates a walkover, signifying that Margaret Court was awarded the championship title without playing the final match due to her opponent, Nancy Richey, withdrawing from the competition.
- In certain years such as 2006, 1990, and 1965, the word "retired" appears in the Score column to indicate that the opponent withdrew during the final set, resulting in an incomplete match.
- In 1977, due to a shift in the tournament calendar, the Australian Open was held twice. The dataset reflects this unique occurrence by labeling them as 1977 (1) for the January edition and 1977 (2) for the December edition.

1.3 Data Structure and Formats

Category	Variable Name(s)	Data Type	Description
Temporal Data	Year	Integer	Year of the final match
Categorical Data	Gender	Categorical	Gender category (Men's or Women's)
	Champion, Runner-up	String (Text)	Names of the players
	Champion Nationality, Runner-up Nationality	Categorical (String)	Nationality codes (ISO format)
	Champion Country, Runner-up Country	Categorical (String)	Full country names
	Score	String	Match score in free-text format
	Champion Seed, Runner-up Seed	Ordinal	Seed rankings of the players (ordinal categorical)
Numerical Data	Mins	Numeric	Duration of the match in minutes
	Games Won, Games Lost	Numeric	Total games won and lost by the champion
	Win Ratio	Numeric (Float)	Proportion of games won out of total games
Set-Level Performance	1st Set Win ratio, 2nd Set Win ratio, 3rd Set Win ratio, 4th Set Win ratio, 5th Set Win ratio	Numeric (Float)	Win ratios for each set (range: 0 to 1)

1.4 Trends Observed

Several trends can be observed from preliminary data inspection:

- **Nationality Trends:** Certain countries like the United States, Australia, and more recently Eastern European nations, have dominated different eras of the tournament.
- **Gender Balance:** Finals for both men's and women's singles have been consistently documented, providing an equal opportunity for gender-based performance analysis.

- **Match Duration:** A general trend of increasing match duration is visible, particularly in men's finals, suggesting higher levels of competition or improvements in athletic endurance.
- **Emergence of Dominant Players:** Some players appear multiple times as champions, indicating the presence of era-defining athletes. These individuals are the focus of the “top players” analysis in the later section of this report.

1.5 Conclusion

This dataset provides a rich foundation for exploring historical performance, trends, and comparative insights across more than a century of Australian Open finals. With well-structured attributes, detailed set-level data, and clearly identifiable outliers, it enables both high-level visual trend discovery and granular player-level analysis. This foundational understanding supports subsequent visualization and clustering efforts to uncover meaningful stories within the data.

II. Data preparation

Data preparation plays a vital role in the data analysis process. Its purpose is to convert raw data into a format that is optimized for analytical tasks. This involves tasks such as detecting and fixing errors, adjusting data types, combining multiple datasets, generating new attributes, or removing irrelevant variables to enhance the effectiveness of data visualization.

- **Missing values:** In the Score column, data is absent for the year 1966 due to a player winning by default without competing. Despite this, the record is preserved to support comprehensive data visualization.
- **Unimportant values:** The variables *Champion Seed*, *Runner-up Seed*, and *Mins* were excluded from the analysis due to their limited relevance to long-term performance trends. *Seed* information, while useful in specific match contexts, does not consistently reflect player dominance across tournaments, and *Mins* (match duration) was inconsistent and incomplete across entries, making it unsuitable for comparative analysis.
- **Boxplot Analysis of Games Won, Games Lost, and Win Ratio**



To identify statistical anomalies and understand the distribution of performance metrics among champions, boxplots were generated for Games Won, Games Lost, and Win Ratio.

Games Won and Games Lost

- The Games Won distribution shows a relatively wide interquartile range (IQR), suggesting diverse performance levels.
- There are two outliers: one significantly lower (below 10 games won) and one higher (over 35 games won), indicating rare match outcomes with either exceptionally quick or extended play.
- The Games Lost distribution is more compact, with lower median and IQR, indicating that most champions lost a limited number of games in their final matches.
- One outlier is detected with more than 30 games lost, which may reflect a highly competitive or extended final.

Win Ratio

- The Win Ratio boxplot shows a centered median around 60%, indicating a balanced average performance across finals.
- One extreme outlier is observed near 0%, likely representing a walkover or retirement match where the player won with minimal play.
- Another outlier near 90% suggests an exceptionally dominant final match performance.

New variables:

Games Won	Games Lost	Win Ratio	1st Set Win ratio	2nd set Win ratio	3rd set Win ratio	4th set Win ratio	5th set Win ratio
15	14	51.72%	66.67%	25.00%	58.33%	0.00%	0.00%
19	12	61.29%	66.67%	53.85%	66.67%	0.00%	0.00%
12	5	70.59%	66.67%	75.00%	0.00%	0.00%	0.00%
24	23	51.06%	33.33%	33.33%	60.00%	60.00%	66.67%
16	13	55.17%	40.00%	66.67%	60.00%	0.00%	0.00%
20	15	57.14%	66.67%	53.85%	53.85%	0.00%	0.00%
13	9	59.09%	66.67%	53.85%	0.00%	0.00%	0.00%
27	26	50.94%	25.00%	46.15%	60.00%	60.00%	58.33%
12	7	63.16%	60.00%	66.67%	0.00%	0.00%	0.00%
19	9	67.86%	58.33%	75.00%	75.00%	0.00%	0.00%

Calculate Games Won and Games Lost

Calculate games won scores as the total number of win points in each set, and Games Lost scores as the total number of loss points in each set.

Calculate Set-Based Win Ratio

The overall win ratio is then calculated as:

$$\text{Win Ratio} = \text{Total Wins} / (\text{Total Wins} + \text{Total Losses}).$$

Calculate the win rate for each set using the formula:

$$\text{1st set WinRate} = \text{1st-won} / (\text{1st-won} + \text{1st-loss})$$

Similarly for the remaining 2 sets in women's matches and 4 sets in men's matches. WinRate = 0% in case there was no set played or the champion did not score any points in the set.

III. Data Visualization

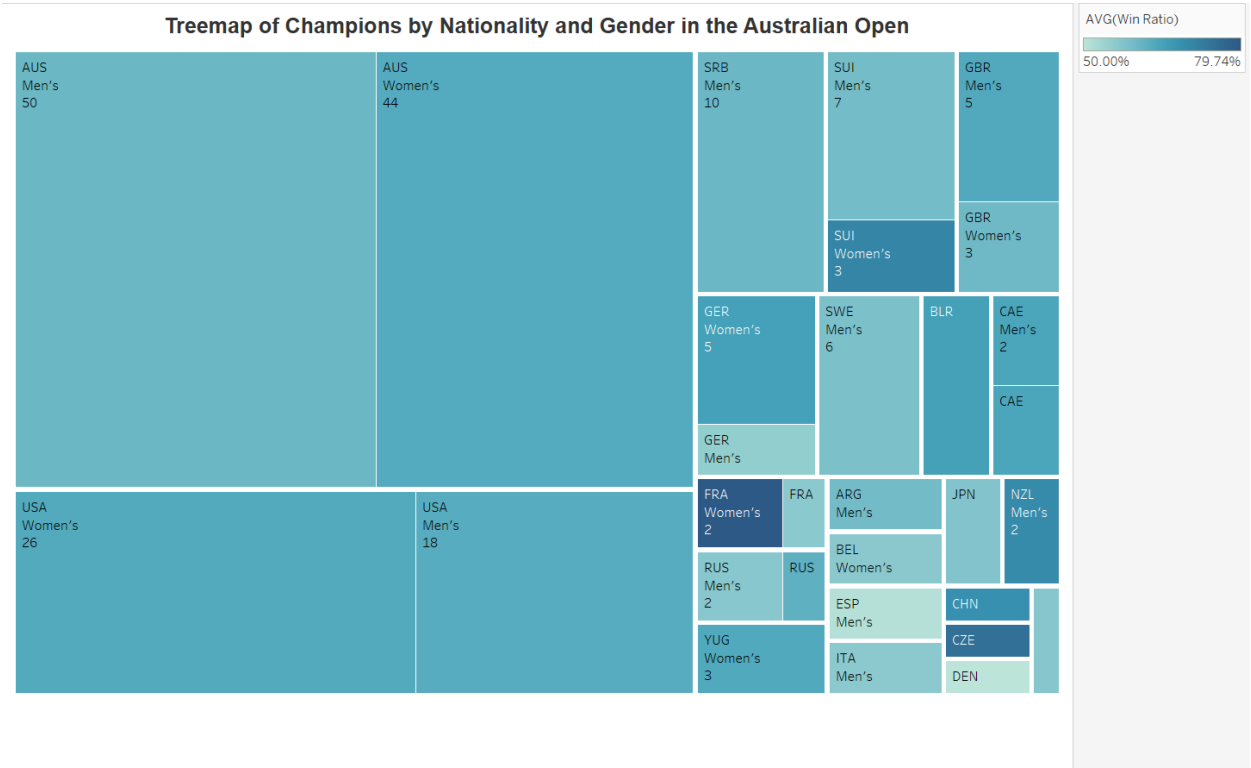
1. Treemap Analysis

The treemap is a powerful visualization technique that displays hierarchical or categorical data using nested rectangles. In this context, each rectangle represents a combination of champion nationality and gender. The size of each rectangle reflects the number of championships won, while the color intensity represents the average win ratio (AVG Win Rate).

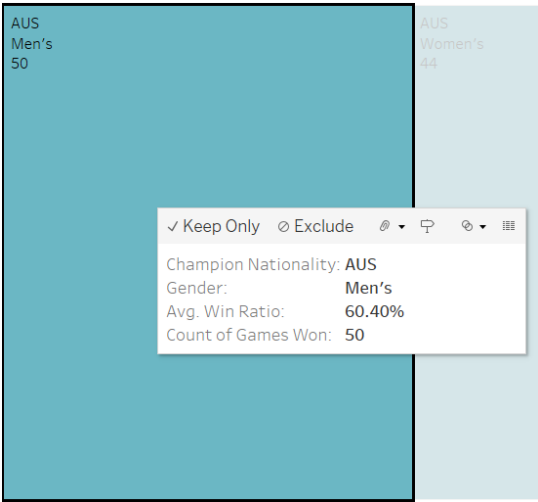
This visualization enables quick identification of dominant countries and gender-based performance trends in the history of the Australian Open.

The treemap implemented in this analysis utilizes a pivoted data structure, enabling hierarchical visualization by champion nationality and gender. Compared to an unpivoted format, the pivoted approach offers superior readability and analytical

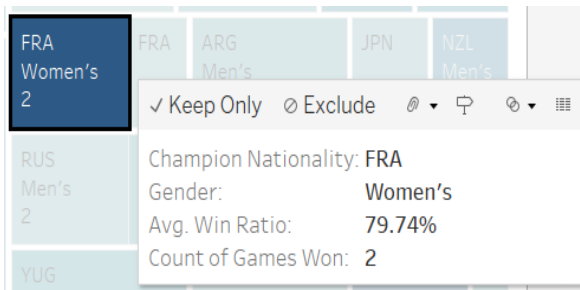
flexibility, allowing for effective comparisons both across and within categories. Therefore, it is a more suitable format for multivariate categorical insights in this championship dataset.



Key Insights:



Australia (AUS) – Men's Division stands out as the most dominant group, occupying the largest rectangle. With 50 total wins and an average win rate of 60.40%, it highlights Australia's strong legacy in men's tennis at the Australian Open. Besides, Australia (AUS) – Women's Division follows closely with 44 titles, reinforcing the country's comprehensive strength in both men's and women's categories.



Although countries like France (FRA) – Women’s Division have significantly fewer titles (only 2 championships), they demonstrate exceptional performance with the highest average win rate of 79.74%. This suggests that although French women’s players did not appear often in the finals, they delivered highly efficient and decisive performances when they did.

This treemap not only reveals the frequency of victories by country and gender but also introduces quality of performance through win ratios. By combining both volume and efficiency, the chart provides a holistic perspective of championship dominance and competitiveness across nations.

Advantages and Disadvantages of Treemap in Data Analysis

Treemap is a highly space-efficient data visualization technique that enables users to represent hierarchical data through nested rectangles. One of its primary advantages lies in its ability to display both proportional size (e.g., number of champions per country) and an additional variable such as average win rate through color gradients. This dual encoding facilitates rapid identification of dominant categories and performance patterns. Furthermore, treemaps are particularly effective in summarizing categorical data with hierarchical dimensions, as demonstrated in the analysis of champions by nationality and gender in the Australian Open dataset.

Despite its strengths, the treemap also presents certain limitations. Firstly, categories with smaller values may be visually minimized, making them difficult to interpret or compare accurately. Secondly, treemaps are less effective when precise numerical values are required, as users must rely on tooltips or legends for detailed information. Lastly, this visualization lacks a temporal component, rendering it unsuitable for trend or time-series analysis. Therefore, while treemaps are powerful for providing an overview of categorical distributions, they are best used in conjunction with other chart types for comprehensive insights.

2. Parallel Coordinates

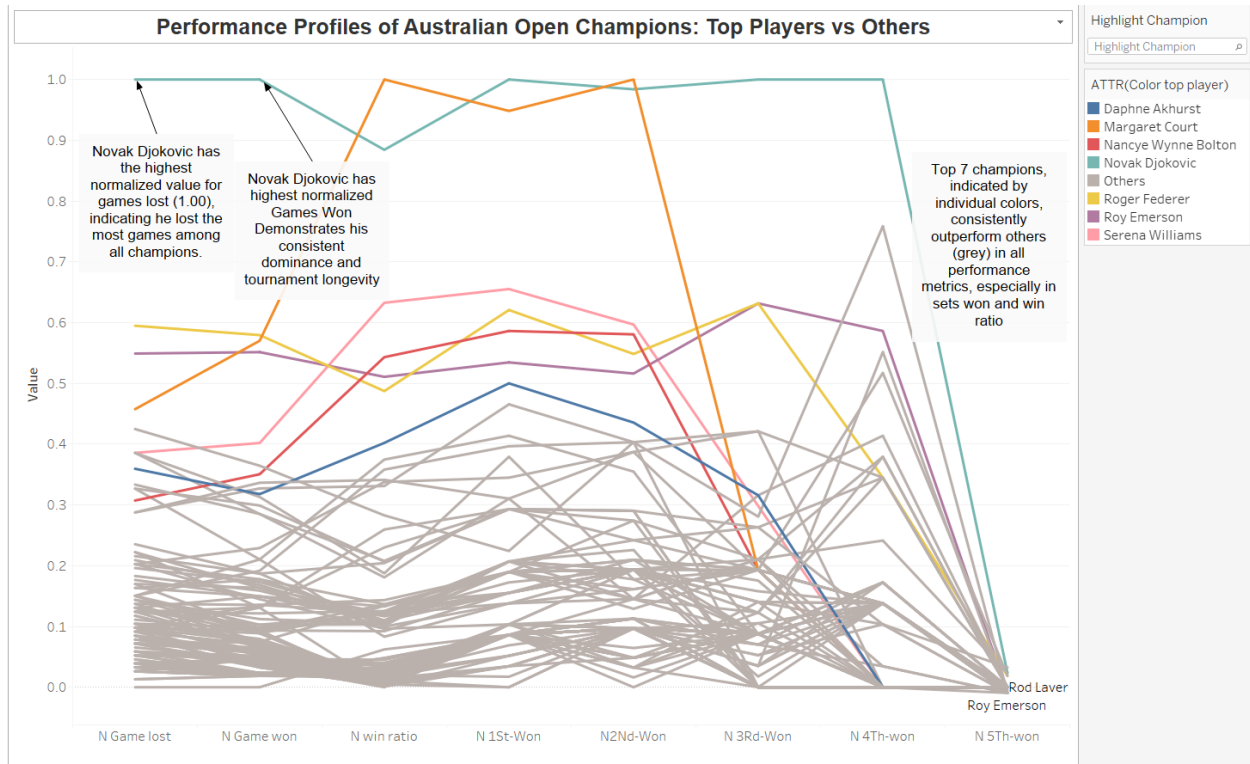
Parallel coordinates are a powerful data visualization technique used to analyze multivariate datasets. Unlike traditional Cartesian plots that are limited to two or three dimensions, parallel coordinates represent each variable as a vertical axis, allowing for the simultaneous visualization of multiple quantitative variables. Each individual observation in the dataset is then represented as a line intersecting each axis at the corresponding value. This technique is particularly effective for identifying trends, patterns, clusters, and outliers in complex datasets.

The parallel coordinates chart is employed to visually compare the performance characteristics of all Australian Open champions across multiple standardized match-related metrics. This chart type was selected due to its effectiveness in displaying multi-dimensional data, allowing readers to identify patterns, deviations, and relationships across several variables simultaneously. To enhance interpretability, top players are defined as those with the highest number of championship titles, assigned distinct colors, while all other champions are displayed in a single neutral color. This design choice emphasizes key outliers and highlights exceptional performance profiles without overcrowding the visual.

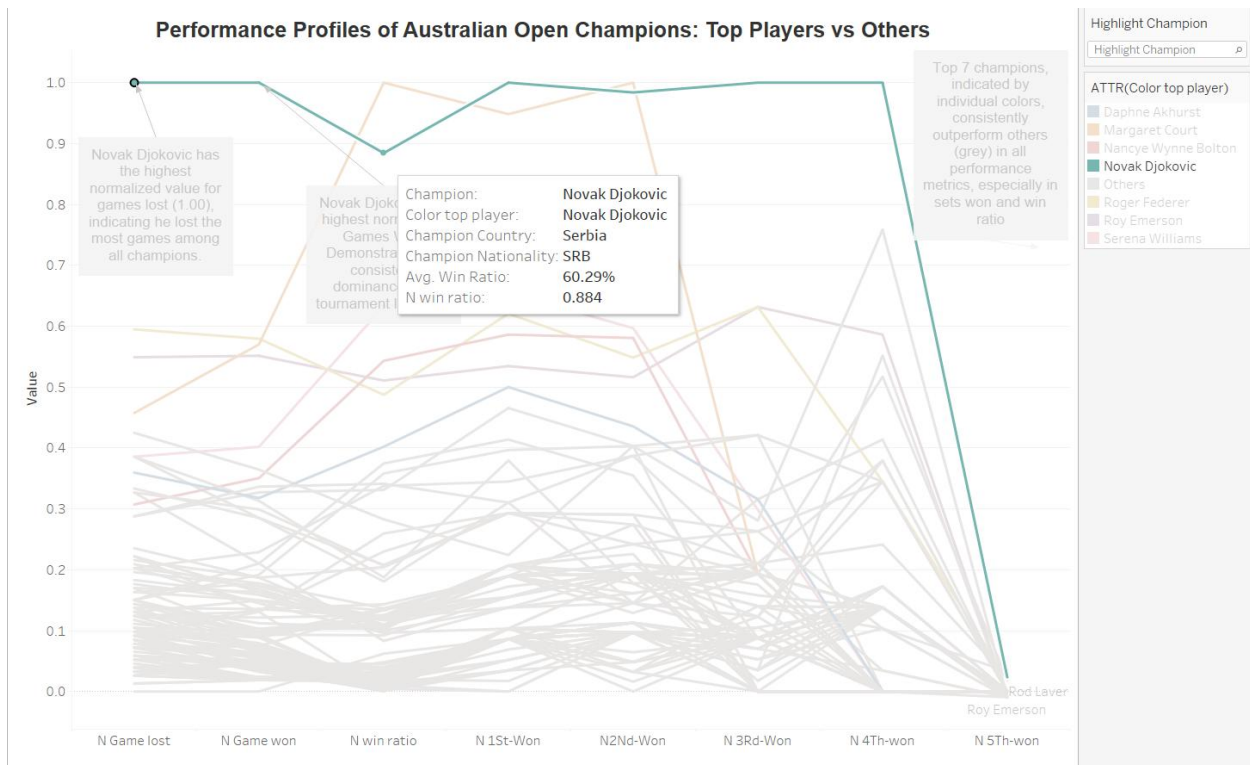
Each axis represents a normalised performance indicator, where “N” stands for “*Normalised*”, meaning the data has been scaled between 0 and 1 to enable fair comparisons across different units and magnitudes. The chart includes:

- N Game Lost: the relative number of games lost, with lower values indicating stronger defensive play.
- N Game Won: the number of games won, where higher values reflect dominance.
- N Win Ratio: the normalized match win percentage.
- N 1st–5th Won: normalized count of set wins in each of the five potential sets, offering insights into match consistency and momentum across early and later stages.

This visualization effectively distinguishes elite players from the broader pool of champions and provides meaningful insights into winning patterns and match performance dynamics among top competitors.



Insights from the Visualization

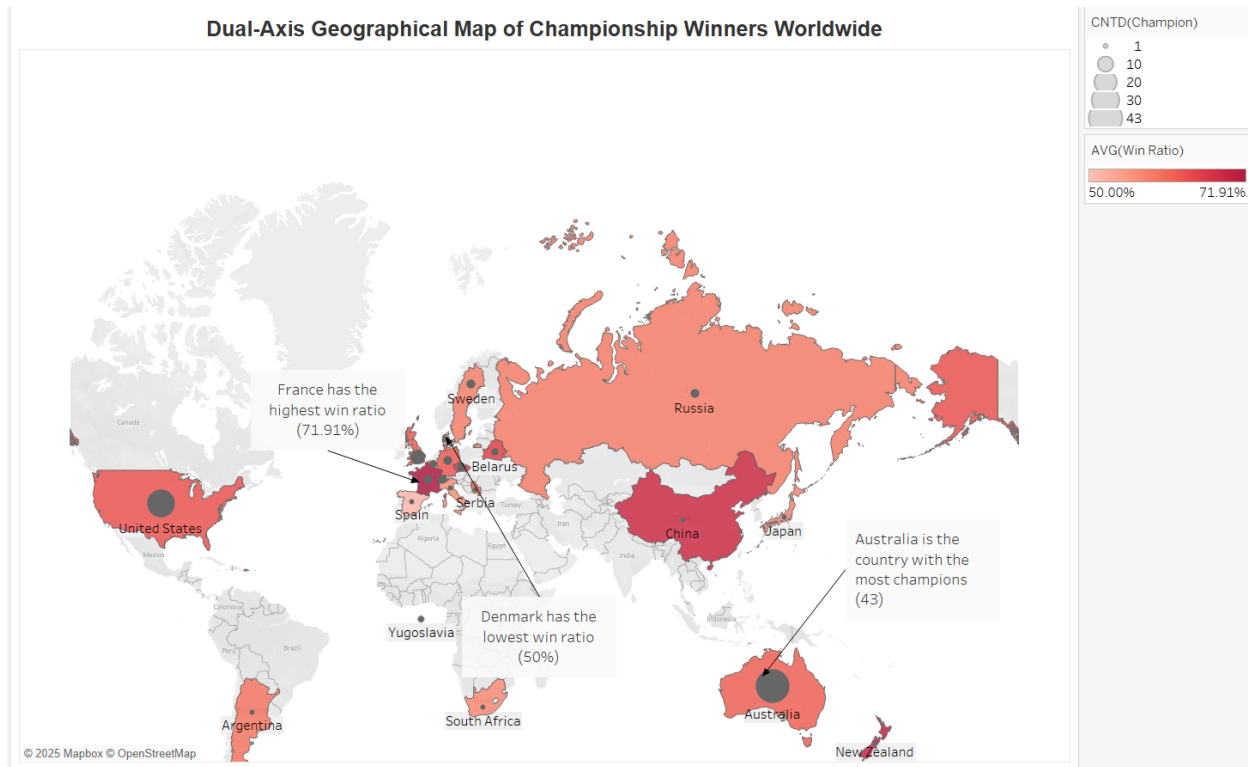


By hovering over or clicking on the teal line representing Novak Djokovic in the parallel coordinates chart, users can observe his exceptional performance across multiple tournament metrics. Compared to the grey lines of other champions, Djokovic's profile remains consistently high, especially in normalized win ratio (0.884) and games won (100%). These indicators reflect not only his dominance in individual matches but also his extensive and successful participation over time. The tooltip interaction reinforces this insight, displaying key statistics such as his win ratio of 60.29% and championship origin. Djokovic's trajectory clearly stands apart from the general trend, making him one of the most consistently high-performing champions in Australian Open history.

The parallel coordinates chart is a powerful visualization technique for comparing multivariate data across multiple individuals. One of its key advantages lies in its ability to display multiple normalized performance metrics simultaneously, enabling a comprehensive comparison of different champions on a single canvas. In this case, the chart effectively distinguishes top players assigned with individual colors from others shown in grey, making it easier to highlight standout performers such as Novak Djokovic. The interactivity of the chart, including hover and tooltip functions, further enhances interpretability by allowing users to explore specific metric values such as normalized games won, win ratio in detail without cluttering the visual space.

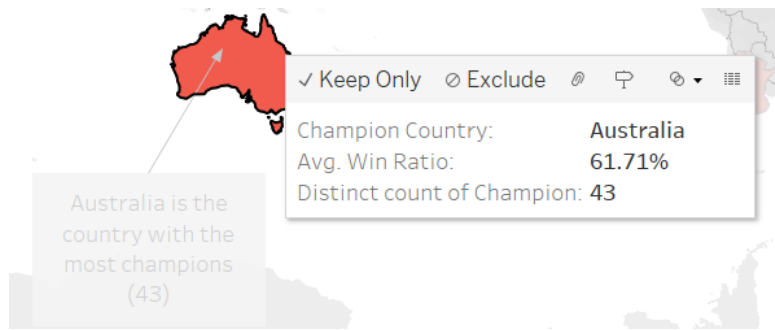
However, the disadvantages include potential visual clutter when too many lines are present, especially for large datasets, which may obscure individual trends among less prominent players. Additionally, interpreting the shape and slope of the lines can become challenging for viewers unfamiliar with parallel coordinates, particularly when the axes are not evenly scaled or labeled intuitively.

3. Geographic map

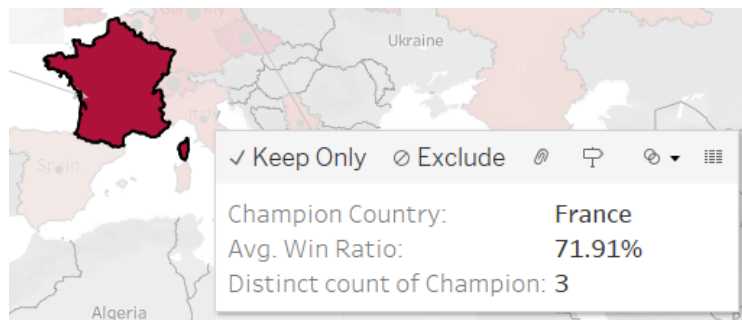


A geographic map is a visualization technique that represents data spatially on a map, using attributes such as size, color, and location to convey insights. It is particularly useful in identifying geographic trends, patterns of distribution, and comparisons across regions. When combined with dual axes as in the dual-axis geographic map, it allows simultaneous visualization of two metrics, enhancing the depth of analysis.

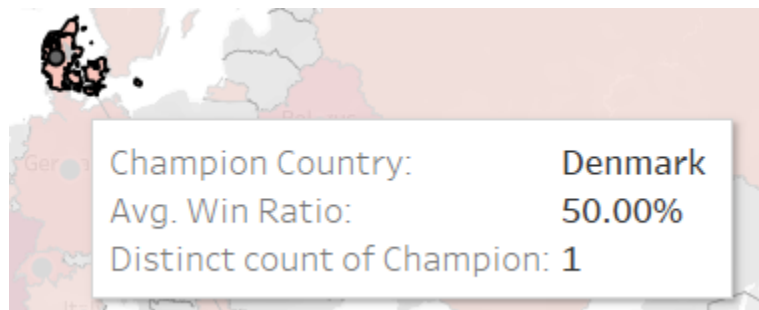
In the context of this analysis, the dual-axis geographic map effectively illustrates the distribution of championship winners across the world while simultaneously highlighting their average win ratios. Here, color intensity represents the average win ratio (from 50% to 71.91%), and circle size indicates the number of unique champions per country. From the visualization:



Australia emerges as the country with the highest number of champions (43), emphasizing its dominance in the Australian Open.



France, although having only two champions, displays the highest average win ratio (71.91%), reflecting exceptional performance in limited appearances.



On the other hand, Denmark has the lowest win ratio (50%), suggesting that while it has representation, its players tend to face more challenges in securing wins.

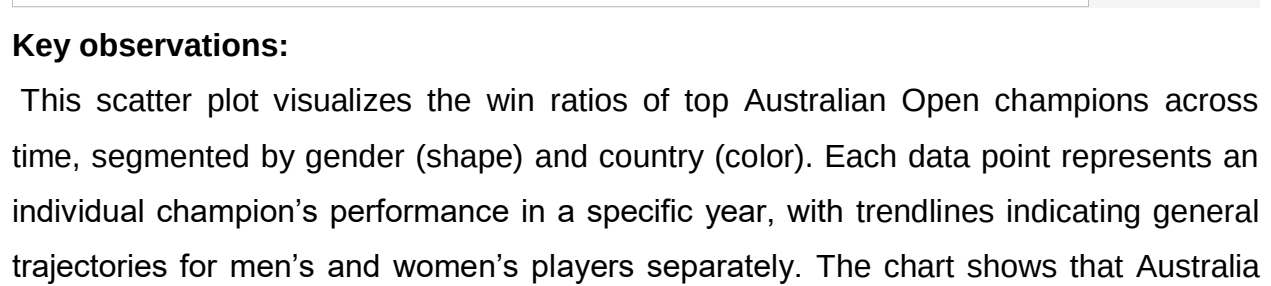
This geographic approach enables a comprehensive view of both participation and performance, offering valuable insights for understanding national-level contributions to the tournament's history.

Advantages and Disadvantages of Dual-Axis Geographic map in Data Analysis

The dual-axis geographic map provides a powerful visual tool for conveying complex spatial data by simultaneously representing quantitative magnitude such as the number of champions through symbol size and performance metrics such as win ratio through color gradients. This dual encoding facilitates comparative analysis across nations, enabling stakeholders to quickly identify high-performing countries and those with the greatest representation. It enhances both interpretability and geographic context, which are essential for evaluating international competition.

4. Scatter chart

The scatter plot visualizes the win ratios of Australian Open champions who have won the tournament at least five times, separated by gender (Men's in green, Women's in orange), across different years. Each dot represents a year a top champion won, annotated with their nationality. The chart includes trend lines to reveal overall performance trends for each gender over time.



(AUS) had several top-performing champions in earlier decades, particularly in the 1960s–1970s. However, in recent years, no Australian players appear among the top champions, as reflected by the absence of AUS-coded points on the right-hand side of the timeline. This suggests a decline in Australia's dominance at the top levels of the Australian Open in modern times.

The analysis reveals several key insights:

- Margaret Court's performance in 1962, marked in orange, stands out with one of the highest win ratios in tournament history (85.71%), reflecting her dominance in women's tennis during that era.
- Conversely, in 1966, Court recorded a 0% win ratio, highlighting an unexpected early exit. This anomaly underscores that even elite athletes may experience performance volatility.
- The trendline for women's players (orange) shows a slightly positive slope, suggesting marginal improvement in win ratios over time, while the men's trendline (green) is relatively flat or slightly declining.
- Recent male champions from Serbia and Switzerland appear clustered around the 55%–65% range, indicating consistent but less dominant performances compared to historic Australian champions.

Advantages and Disadvantages of the Chart

The scatter plot offers an effective visualization of longitudinal trends in win ratios among top Australian Open champions. By encoding country with color and gender with shape, the chart supports multidimensional analysis while maintaining visual clarity. The inclusion of annotations further enhances interpretability by highlighting significant historical events, such as Margaret Court's peak performance and anomalous results. However, the representation of players with multiple participations as individual data points may result in visual redundancy, and extreme outliers can affect overall interpretability if not contextualized appropriately.

IV. Executive summary and Conclusion

1. Executive Summary

This report analyzes over 120 years of Australian Open championship data using advanced data visualisation techniques to uncover patterns, performance trends, and national dominance in both men's and women's singles. Through a series of well-designed visualisations that including treemap, parallel coordinates, geographic maps, and scatter plots. The analysis reveals the evolution of champion profiles over time and highlights standout players and nations.

The treemap revealed Australia's historical dominance in championship wins, while France, despite fewer appearances, demonstrated the highest average win ratios. Geographic mapping supported these findings, showing Australia's early lead but an increasing diversity of champion origins in recent decades.

Parallel coordinates enabled detailed player-to-player comparison, with Novak Djokovic showing remarkable consistency and performance across multiple metrics. The scatter plot further demonstrated that Australia, once a dominant force, has not produced top champions in recent decades, reflecting a shift in global tennis competitiveness.

The report applies effective storytelling through clear labeling, use of color, gender differentiation, and annotations to highlight key moments such as Margaret Court's career highs and anomalies. Tableau's visualisation tools were instrumental in enabling interactive filtering, layering, and trend identification to support both macro and micro-level insights.

2. Conclusion

The visual analysis of the Australian Open champions presents a compelling narrative of performance evolution and shifting dominance over time. Australia produced many top-tier champions during the mid-20th century, especially in women's tennis, but in recent years, no Australian players have appeared among the highest-performing champions. This reflects a decline in local dominance and a growing presence of international champions, particularly from nations like Serbia and Switzerland.

Gender-based trendlines reveal a marginal upward trend in win ratios for women, whereas men's performances appear more stable over time. Comparative visualisations also highlight individual excellence, with players such as Novak Djokovic and Margaret Court demonstrating sustained dominance.

Tableau's rich suite of features, including dual-axis plotting, shape and color encoding, dynamic filters, and annotation tools that proved invaluable for turning complex, multidimensional data into accessible, interpretable insights. These visualisations not only enhanced readability but also facilitated deeper comparative analysis, making the report both informative and engaging.